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AI_{Milano} Apoprotein Identification of the Complete Kindred and Evidence of a Dominant Genetic Transmission

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SUMMARY

The AI_{Milano} apoprotein variant is associated with a marked reduction of high density lipoprotein (HDL) cholesterol levels and with increased triglyceridemia. In spite of the low HDL-cholesterol (HDL-Ch), carriers do not generally show clinical signs of atherosclerosis. The biochemical disorder is linked to a molecular change in apoprotein AI, that is, an arg → cys substitution in the 173 position, thus allowing the formation of AI_{Milano}-AI_{Milano} dimers and AI_{Milano}-AII complexes. The origin of the variant gene has been located in Limone sul Garda, a small community in Northern Italy (about 1,000 individuals). This community has a genetic, biochemical, and clinical individuality, consequent to its isolation up to a few years ago; the citizens show highly uniform alimentary habits and elevated consanguinity.

The complete population of the small village was sampled, and, by the use of an analytical isoelectric focusing technique for the detection of the mutant, a total of 33 living carriers, ranging in age from 2 to 81 yrs, were identified. Analysis of the genealogic tree of the complete family groups showed that the apoprotein (apo) AI_{Milano} is transmitted as an autosomal dominant trait, all carriers coming from a single mating couple, living in the eighteenth century. The carriers are heterozygous for the apoprotein variant.

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Lipid-lipoprotein parameters of the AI_{Milano} carriers were compared with the corresponding sex and age classes of the general population. A significant reduction in the HDL-Ch levels and a considerable mean increase of triglyceridemia were detected, the latter finding not reaching statistical significance because of the small number of AI_{Milano}-positive individuals. These represent the most numerous known group of carriers of an apolipoprotein mutation and, from the clinical point of view, show an unusually low prevalence of atherosclerotic vascular conditions.

INTRODUCTION

The AI_{Milano} apoprotein was the first described amino acid variant of human apolipoproteins [1, 2]. Carriers of the mutant apoprotein show a marked decrease of high density lipoprotein cholesterol (HDL-Ch) levels and increased plasma triglycerides (TG). In spite of the reduction of HDL-Ch, they do not generally show any clinical sign of atherosclerotic disease. The biochemical basis of this metabolic disorder has been linked to the presence of a well-defined molecular change in apoprotein (apo) AI: the AI_{Milano} (AI_M) is, in fact, characterized by a single cysteine for arginine replacement at position 173 of apo AI [3]. This amino acid substitution accounts for the charge difference between variant and normal AI isoforms detected by analytical isoelectric focusing (IEF) [4].

A significant variability in the abnormal apo AI content was detected in the five subjects, in whom separation of the two forms was carried out [3, 4]. The statistical correlation between the quantitative presence of the mutant and phenotypic expression of the above-described abnormal metabolic patter [4] prompted considerable interest in the genetic and/or metabolic regulation of plasma levels of both the normal and variant apo AI in the AI_M carriers.

Since the origin of the variant gene has been located in Limone sul Garda, a small community in Northern Italy, the complete population of the village was sampled in order to identify as many as possible families carrying the molecular mutant. The pedigrees of these families were analyzed in order to: (1) study the segregation and verify the Mendelian model of inheritance; (2) completely explore the phenotype, both clinical and biochemical, which is individually linked to the presence of apo AI_M; and (3) evaluate whether the mutant phenotype may interfere in the mechanism of natural selection, within a closed and well-defined community.

MATERIALS AND METHODS

Subjects

The major clinical and biochemical findings of the original AI_M proband and of his close relatives have been reported [1–4]. The family originated from Limone sul Garda, a small village on the Garda Lake, in Northern Italy. In January 1982, the population

was invited to participate in a clinical study aimed to the search of new carriers of the mutant apoprotein.

Eight hundred fifteen subjects, that is, 94% of all citizens older than age 10, were examined. The genetic and biochemical characteristics of this population, excluding the AI_M carriers, have been reported [5].

Protocol of the Population Study

During the first visit, samples of 10 ml of blood were collected from all the 815 participating subjects after an overnight fast. The subjects were weighed, and their sitting blood pressure determined after 15 min of rest. All provided information on personal and family health histories. Carriers of the AI_M and their close relatives, following identification, as described below, were re-examined after 6 and 9 months, with a complete health screening, inclusive of general medical observations (arterial auscultation, eye ground, ECG) and a standard battery of hematological and biochemical tests.

Mortality data for the last 10 years (1972–1981) for the whole population were, at the same time, collected by the Secretariat of the City Hall. As a reference, the age-sex distribution for the whole Italian population in 1976 (an intermediary year along the considered period) was used. The 81 deaths, observed during the 10-year period, were classified into six nosographic groups according to the 8th revision of the WHO-ICD codes [6] and the expected deaths computed from the Italian 5-year age-specific rates (1976) [7], applied to the age distribution of the Limone population (table 1). Methods for obtaining pedigree information and calculating consanguinity indexes have been reported [5].

Biochemical Tests

Plasmas were obtained by low-speed centrifugation within 3 hrs from sampling and processed for cholesterol (Ch) [8], TG [9], and HDL-Ch [10] within 24 hrs. Plasma total Ch, TG, and HDL-Ch determinations are standardized within an Italian Collaborative Study [11], under WHO (Prague Laboratory) Quality Control System supervision.

TABLE 1

DEATHS OBSERVED IN LIMONE SUL GARDA IN A 10-YR PERIOD (1972–1981) CLASSIFIED INTO SIX NOSOGRAPHIC GROUPS ACCORDING TO THE REVISION OF THE WHO-ICD CODES [6] AND EXPECTED DEATHS COMPUTED FROM THE ITALIAN 5-YR AGE-SPECIFIC RATE (IN 1976) APPLIED TO THE AGE DISTRIBUTION OF THE LIMONE POPULATION

Causes of death	Observed O	Expected E	O minus E as % of E	P of χ^2
Coronary heart disease—codes 410–414	7	14.32	−51.1	n.s. ($.10 > P > .05$)
Strokes—codes 430–438	12	12.51	−4.1	n.s.
Other cardiovascular diseases—codes (390–458) minus codes (410–414) minus codes (430–438)	16	13.99	+14.4	n.s.
Cancer—codes 140–239	18	17.76	+1.3	n.s.
Violence—codes 800–999	4	4.87	−18.0	n.s.
Other causes not mentioned above	24	24.97	−3.9	n.s.
All causes of death—codes 000–999	81	88.45	−8.4	n.s.

The AI_M carriers were identified after precipitation of the apo B-containing lipoproteins by the dextran-$MgCl_2$ procedure [10]. Added were 0.5 ml of the supernatants with $MnCl_2$ (0.2 M final) and the dextran sulphate concentration raised to 0.5% (w/w) in order to precipitate HDL [12]. After an overnight stay at 4°C, the precipitates were washed with 0.09 M $MnCl_2$ and 0.09% dextran sulphate and resuspended in 0.5 ml of 0.09 M Na_2CO_3. Fifty μl of the clear supernatants were delipidated with chloroform/methanol (2/1) and stored at −20°C; analyses were performed within 2 weeks.

IEF slab gels (160 × 140 mm, 1.5-mm thick) were prepared in 7.5% acrylamide, 8 M urea with 2% ampholytes (Ampholine, pH 4–6, LKB, Bromma, Sweden) and cast on a "Protean Double Slab Cell" (Bio Rad, Richmond, Calif.). After a prerun of 30 min at 200 V, 18 samples (dissolved in 100 μl of 8 M urea, 10% sucrose) were applied to each gel and focalized for 16–18 hrs at 400 V. Gels were fixed and stained with Coomassie Blue according to standard procedures [13].

Statistical Methods

To compare the means of the biochemical findings between the apo AI_M carriers and the general population, corresponding for age and sex, the C test (normal deviate) was used. This test, also called t', is recommended by Snedecor and Cochrane [14], when the compared means refer to samples of different size and with a different variance. In the calculation of the interlipid correlations, we have used individual data of biochemical parameters, compared to the mean and the standard deviation of the corresponding age class. The formula to transform the laboratory data (X_{lab}) to those used for calculation (X_{cal}) is: $X_{cal} = (X_{lab} - m)/s$, where m = mean and s = SD.

The plasma lipid correlations were studied by calculating the r correlation indexes and by estimating the significance of deviations from 1 with the t-test. The correlations detected in the AI_M carriers were compared with those of the general population by using the respective correlation indexes, setting the null hypothesis that the different indexes (r_s) are an estimate of a same index (e) characteristic of a single population: the statistical method employed the χ^2 test as suggested by Snedecor and Cochrane [14].

RESULTS

Prevalence of the AI_{Milano} Carriers and Formal Genetic Analysis

By the use of the above-described analytical IEF technique (fig. 1), 28 subjects were identified as carriers of the apoprotein variant AI_M, thus bringing the total number of known carriers to 33 (table 2). The genealogic tree, reported in figure 2, is comprehensive of all identified cases, who, therefore, share a common genetic origin. The mutation, in the examined population sample, goes back to one of the two mates of the I generation, living in the second half of the eighteenth century.

The global observation of the segregatory features of the mutated trait through the generations allows the following conclusions: (1) the transmission is direct without any generational jump; and (2) the phenotype has the same frequency in the two sexes ($\chi^2 = 1.336$).

The transmission of apo AI_M takes place according to the Mendelian model of autosomic dominant inheritance. All the observed crosses may be identified according to the model $Aa \times aa$. By testing the theoretical data with the observed findings, a $\chi^2 = 0.336$ may be obtained, thus allowing to accept the hypothesis of a dominant autosomic inheritance (table 3).

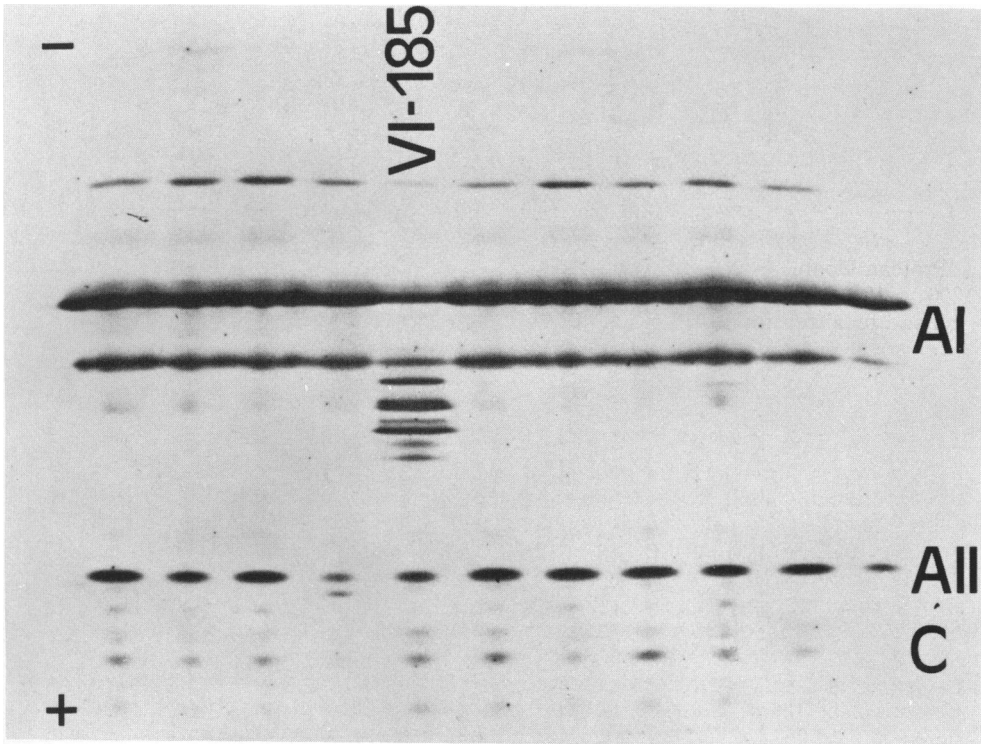


FIG. 1.—IEF gel (pH range 4.0–6.0) of HDL apoproteins isolated by sequential precipitation from 11 subjects living in Limone sul Garda. The abnormal pattern due to the AI_M variant has been detected in subject VI-185, thus identified as a carrier of the mutant apoprotein.

Biochemical Findings

In table 4, the biochemical findings in the AI_M subjects, compared to those of the general population [5], taking into account age and sex, are presented. Carriers have a lower mean total Ch, not, however, significantly different from the general population. On the other hand, the relative hypocholesterolemia of the AI_M carriers is related to the very marked reduction of the HDL-Ch levels: this difference is statistically significant. The same AI_M carriers also show a clear hypertriglyceridemia, compared to the rest of the population, but again a level of statistical significance is not achieved, by the C test, due to the small number of carriers.

The correlation indexes between the plasma lipid/lipoprotein levels in the AI_M carriers and in the general population have also been calculated and are reported in table 5. We have, moreover, compared the lipid correlation indexes of the carriers with those of the general population of Limone sul Garda [5]. The following results were obtained:

Ch/HDL-Ch	AI_M carriers	$r = -.0115$ (no. = 33)
comparison	General population	$r = .248$ (no. = 671)
		$\chi^2 = 4.464$

TABLE 2

AGE, SEX, BLOOD GROUPS, BLOOD PRESSURE, AND BIOCHEMICAL AND CLINICAL CHARACTERISTICS OF THE AI_M CARRIERS

Kindred position	Age	Blood group	P-CH	P-TG	HDL-Ch	BMI	SBP	DBP	HLP type	Clinical observations
						Males				
V-25	81	O+	183	225	12	28.69	145	80	IV	Varicose veins
V-84	75	B−	187	243	18	23.74	165	90	IV	Atrial fibrillation, duodenal ulcer, apical systolic murmur, hepatomegaly
V-88	69	AB−	196	151	25	22.15	200	120	...	Aortoiliac bypass, peripheral arterial disease, left ventricle enlargement
VI-21	55	O+	154	199	9	25.88	120	80	...	Silicosis
VI-58	50	O−	298	366	6	26.99	130	75	IV	Tonsillectomy, duodenal ulcer, hepatomegaly
VI-62	39	A+	270	230	12	23.84	130	90	IV	...
VI-171	39	A+	210	262	10	30.61	140	80	IV	...
VI-190	34	B−	152	82	23	22.49	100	60	...	Varicose veins, incomplete right bundle, branch block
VI-205	47	A+	233	273	14	26.61	140	90	IV	Hepatomegaly, femoral bruit
VI-207	40	O+	279	258	14	24.91	125	80	IV	...
VII-46	21	O+	135	146	12	23.81	120	80	...	Tonsillectomy, incomplete right bundle branch block
VII-84	21	O+	193	261	9	20.37	135	85	IV	Tonsillectomy
VII-134	29	A−	153	107	14	22.21	140	80	...	Tonsillectomy, hyperbilirubinemia
VII-136	17	A−	125	95	20	22.39	140	80	...	Tonsillectomy, hyperbilirubinemia
VII-137	17	A+	156	160	7	21.97	140	80	...	Tonsillectomy, hyperbilirubinemia
VII-153	18	O+	206	135	31	21.91	130	80	...	Carotid bruit
VII-171	13	O+	147	77	13	18.67	100	70	...	...
VII-192	2	A+	180	107	13	...	...	...	...	...
VII-207	9	A+	143	92	14	21.17	110	60	...	...
VII-208	8	A−	180	94	9	16.33	95	70	...	...
m			189.00	178.15	14.25	...	...	...	...	...
s			48.86	82.42	6.32	...	...	...	...	...

Females										
VI-91	65	B−	218	162	35	27.56	150	95		Tonsillectomy, appendectomy, ing. hernia, pleuritis, arthrosis
VI-126	53	A+	237	329	17	30.86	180	110	IV	Tonsillectomy, pleuritis, hypertension (eye ground KWI)
VI-132	39	O+	172	172	34	23.51	140	90	...	Appendectomy, varicose veins
VI-176	35	A+	162	110	24	23.71	120	70	...	Tonsillectomy, appendectomy, polio
VI-185	27	B+	172	188	25	20.20	120	80	...	Pleuritis, tonsillectomy
VI-196	32	A−	185	243	12	23.89	120	80	IV	Varicose veins, altered intraventricular conduction
VI-203	60	A+	253	228	19	22.35	120	80	IV	Discal hernia, xanthelasma, varicose veins
VII-48	13	O+	139	61	17	21.93	90	61	...	Tonsillectomy, appendectomy
VII-86	14	A+	196	178	16	21.83	100	75	...	Tonsillectomy
VII-149	31	A+	215	277	14	26.49	120	80	IV	Varicose veins, carotid bruit
VII-178	2	A+	152	156	10	...	...	...	...	...
VII-201	22	A+	185	151	24	24.46	120	80	...	Varicose veins
VII-206	15	O+	138	80	21	19.61	100	70	...	Appendectomy
m			186.46	179.62	20.62	...	...	...	...	...
s			36.16	75.73	7.69	...	...	...	...	...
Total sample										
m			188.00	178.73	16.76	...	...	...	...	...
s			43.69	78.64	7.46	...	...	...	...	...

NOTE: P-CH = plasma total cholesterol; P-TG = plasma total triglycerides; HDL-Ch = high-density lipoprotein cholesterol; BMI = body mass index (weight, kg/height, m^2); SBP = systolic blood pressure; DBP = diastolic blood pressure; HLP = hyperlipoproteinemia; m = mean; s = SD.

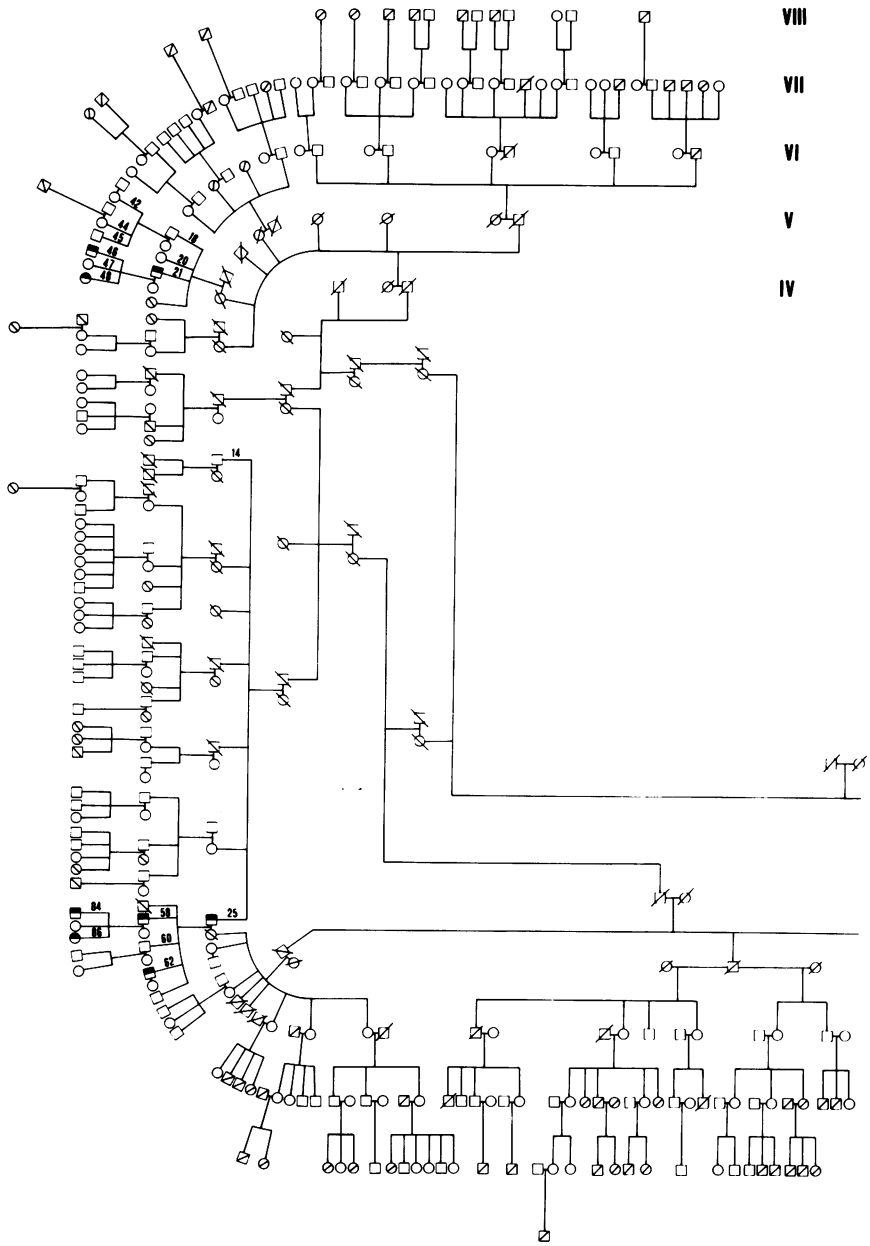


FIG. 2.—Legend appears on next page

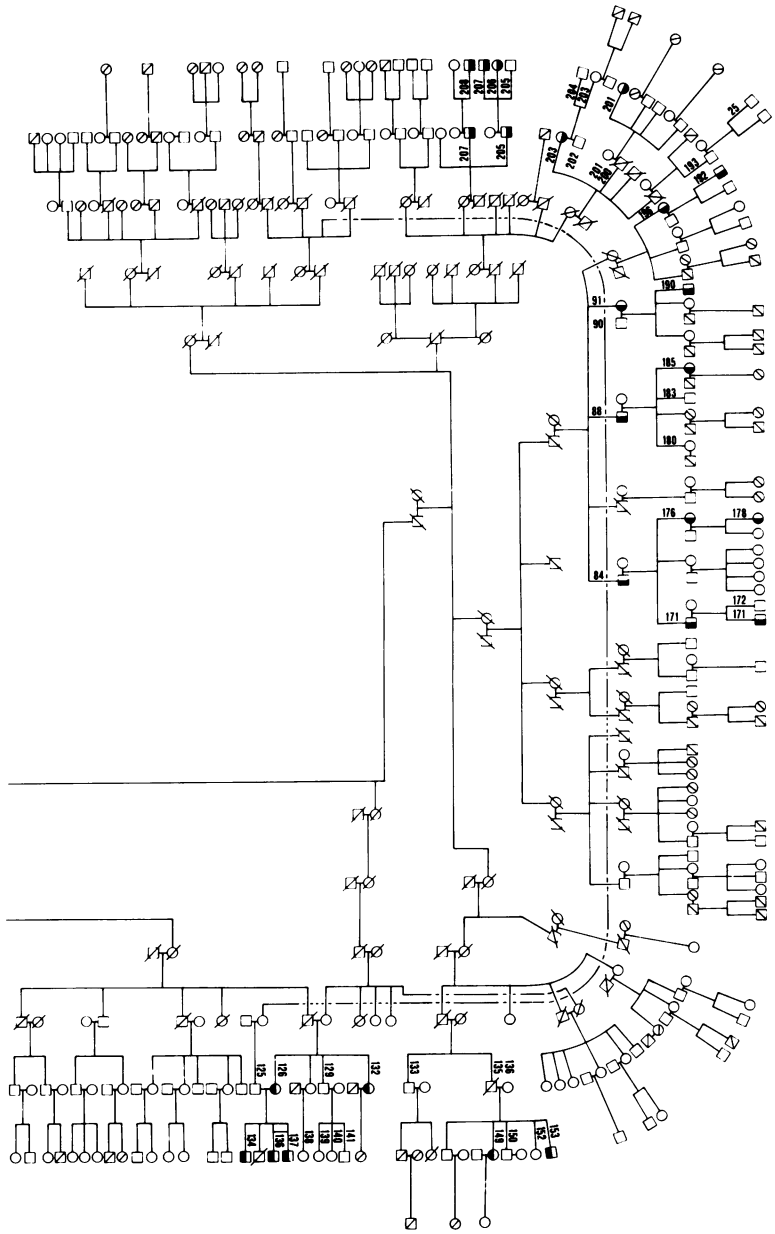


FIG. 2.—Pedigree of the complete AI_M kindred. Symbols ◧◐, □○, ⊘⊘, ⊘⊘ identify: AI_M carriers, noncarriers, noninvestigated subjects, and deceased individuals, respectively, males and females. The original proband, subject D. V. [1], corresponds to subject VI-58.

TABLE 3

GENETIC ANALYSIS OF THE TRANSMISSION OF THE AI_M TRAIT IN THE DIFFERENT ANALYZED SIBSHIPS

Size of sibship/s	No. sibships	No. offspring	AI_M^- males	AI_M^- females	AI_M^+ males	AI_M^+ females
2	2	4	1	1	2	0
3	5	15	2	4	4	5
4	1	4	2	0	2	0
Total	8	23	5	5	8	5
Crosses: AI_M^- male × AI_M^+ female						
1	1	1	0	1	0	0
2	3	6	2	2	1	1
3	2	6	0	2	4	0
Total	6	13	2	5	5	1
Grand total	14	36	7	10	13	6

Ch/TG comparison	AI_M carriers	r = .789 (no. = 33)
	General population	r = .327 (no. = 671)
		χ^2 = 14.618
TG/HDL-Ch comparison	AI_M carriers	r = .336 (no. = 33)
	General population	r = .250 (no. = 671)
		χ^2 = .475

General Clinical Observations

The general clinical findings in the affected carriers, compared to close relatives, were, altogether, unremarkable (table 6). Only one of the 33 carriers had, in fact, a definite evidence of atherosclerotic lesions, as witnessed by the necessity of an aorto-femoral bypass (V-88). This subject has an advanced age and has one of the highest HDL-cholesterolemias noted among the AI_M carriers (table 2). Two other deaths may have been attributable to atherosclerotic complications in obligate carriers in recent years. Subjects VI-200 and VI-135 died, in fact, at relatively young ages (respectively, 59 and 54): in the former case, because of a "heart attack"; in the latter, of cerebrovascular disease. The autopsy report of case VI-200 shows that the patient had a perfectly normal coronary vasculature, the cause of death being tuberculous pericarditis. In case VI-135, although an autopsy was not carried out, the clinical chart is sufficiently informative: the patient was a heavy smoker with arterial hypertension, his ultimate cause of death being cerebral hemorrhage. Two young carriers had clear carotid bruits (VII-149 and VII-153). By B-mode sonography (Biosound), in both cases, carotid arteries were found to be morphologically normal (fig. 3).

TABLE 4

MEAN AND DISPERSION OF PLASMA CH, TG, AND HDL-CH IN THE APO AI_M CARRIERS: COMPARISON WITH THE CORRESPONDING CLASSES OF THE GENERAL POPULATION

	TOTAL CH			TG			HDL-CH		
SUBJECTS	m	s	C	m	s	C	m	s	C
Males < 20 yrs									
AI_M	162.4	27.60	...	108.57	28.91	...	15.29	8.06	...
Limone	175.3	29.50	...	65.50	29.60	...	43.90	9.30	...
Males > 20 yrs									
AI_M	203.3	52.60	...	215.62	77.41	...	13.46	4.99	...
Limone	237.2	44.00	...	133.57	76.44	...	46.83	11.30	...
All males									
AI_M	189.0	48.85	...	178.15	82.42	...	14.25	6.32	...
Limone	229.4	42.20	0.802	125.06	70.59	0.304	46.46	11.05	2.199*
Females < 20 yrs									
AI_M	156.3	27.30	...	118.75	56.96	...	16.00	4.55	...
Limone	177.4	29.30	...	71.90	30.60	...	45.80	8.10	...
Females > 20 yrs									
AI_M	199.9	31.90	...	206.67	68.63	...	22.66	8.08	...
Limone	238.8	27.78	...	116.39	53.63	...	50.21	8.86	...
All females									
AI_M	186.5	36.16	...	179.62	75.73	...	20.62	7.69	...
Limone	231.1	33.65	0.412	110.82	50.75	0.884	49.66	9.64	2.604*

NOTE: C = Normal deviate test: comparison with the mean of classes in the general population, corresponding for age and sex [5]; m = mean; s = SD.
* $P = .01$.

ECG findings, as assessed according to the Minnesota Code [15] were, altogether, of minor significance, with the exception of a few cases of abnormal repolarization in the close relatives.

DISCUSSION

Our study provides detailed genetic and clinical information on the carriers of the AI_M mutant in the small community where, apparently, the variant apoprotein had originated. All the described carriers are descendants of the original mating couple. This observation does not, of course, rule out the possibility that either of the spouses had acquired the trait from relatives living in neighboring communities. Limone sul Garda proved to be a village of unusually low cardiovascular morbidity and mortality. These observations should not be attributed to the presence of an apparently protective gene, in view of the relatively low percentage of carriers. The examination of this subsample, consisting now of 33 subjects, of various ages and sex, confirms that the apoprotein abnormality does not seem to be linked to any significant risk of cardiovascular disease. Moreover, differently from other conditions associated with HDL abnormalities (e.g., fish-eye disease [16] and Tangier disease [17,

TABLE 5

CORRELATION INDEXES BETWEEN THE DIFFERENT LIPID-LIPOPROTEIN PARAMETERS FOR THE AI_M CARRIERS

Comparisons	No.	r	t	P
TCh/HDL-Ch				
Males	20	−.149	0.685	.60–.50
Females	13	−.073	0.243	.90–.80
All	33	−.115	0.645	.60–.50
TCh/TG				
Males	20	.777	5.241	< .001
Females	13	.801	4.434	< .001
All	33	.789	7.153	< .001
HDL-Ch/TG				
Males	20	−.394	1.819	.10–.05
Females	13	−.257	0.882	.40–.30
All	33	−.336	1.988	.10–.05

18]), there does not appear to be any clear association between the apoprotein abnormality and any pathological condition, particularly cardiovascular.

The phenotype of the apo AI_M mutation has been, up to now, described in the heterozygote state, and we have not acquired knowledge of the phenotype corresponding to the homozygote. From examining the genealogic tree, it ap-

TABLE 6

MAJOR CLINICAL OBSERVATIONS IN THE AI_M CARRIERS (NO. = 33) AND IN THEIR CLOSE RELATIVES (NO. = 43)

	AI_{Milano} carriers	Close relatives
Appendectomy	5	10
Arterial bypass procedure	1	...
Arthrosis	1	3
Atherosclerotic peripheral	1	1
Arterial disease	...	...
Cardiac arrythmias	1	...
Discal hernia	1	1
Gallstone disease	...	2
Gastroduodenitis—peptic ulcer	2	1
Eye ground vascular abnormalities	1	5
Heart murmurs	2	1
Hemiparesis	...	1
Hepatomegaly	4	4
Hyperbilirubinemia	3	...
Hypertension	1	5
Inguinal hernia	1	4
Peripheral arterial bruits	3	1
Pleuritis	3	1
Poliomyelitis	1	1
Silicosis	1	...
Thyroid dysfunction	...	1
Tonsillectomy	12	10
Tonsillitis	...	2
Venous disease	7	5
Xanthelasma	1	1

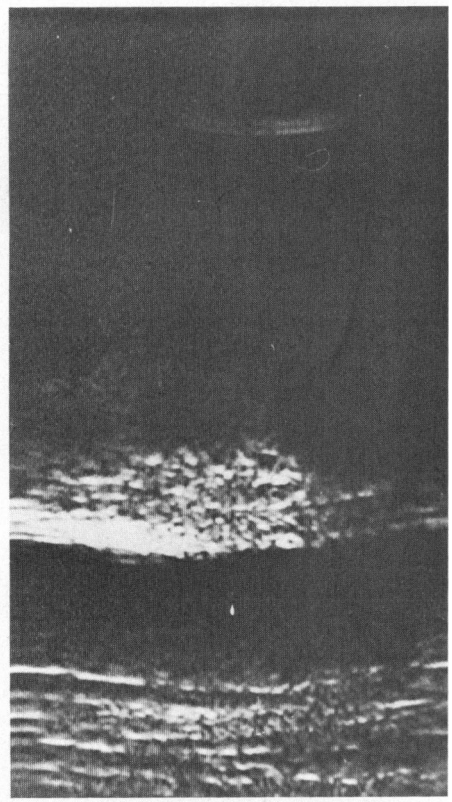


FIG. 3.—Carotid high-resolution real-time sonography (B-mode, Biosound) in a carrier of the AI_M variant, presenting with a significant carotid bruit (VII-149). Both the anterior and posterior walls of the artery (*left* and *right*) appear perfectly normal and free of plaques.

pears that genetic conditions predicting the appearance of homozygotes were never present. In the case of other inherited abnormalities of lipoprotein metabolism (e.g., genetic Tangier disease [17], fish-eye disease [16], and the apo AI-CIII deficiency [19]), the characteristic phenotypic traits, both biochemical and clinical, are, instead, perceptible only in the homozygotes.

By examining the large genealogic tree of figure 2, and by trying to pick out probable carriers, based on the segregation of the trait and the familial composition, we have been able to note that these had a relatively long life, compared to the mean for the place and period (1800). A question we have raised and on which we will work in the future refers to the interference of environmental conditions on the apo AI mutation, that is, on the activity of natural selection in favor or against the mutant. The follow-up of carriers, from now on exposed to the modern atherogenic risk, will indicate if the apo AI mutation really offers a selective advantage.

Some biochemical findings, associated with the classically reported HDL deficiency, were of interest among the studied subjects. In general, mild-to-moderate hypertriglyceridemia, with total and low density lipoprotein cholesterolemia in the normal range, were detected. A rise of triglyceridemia with age

was also noted, nonsignficant because of the small number of probands. A complex mechanism, associating reduced HDL lipids and proteins with an impaired TG catabolism, may be present in the AI_M carriers [20].

The phenotypic expression of the AI_M mutation is variable [3], as supported by the large spread of the HDL-Ch data, as well as of other associated biochemical abnormalities. From data obtained in a small number of carriers, it appears that HDL-Ch levels are directly related to the amount of normal AI [4]. The AI_M variant originated in a community with a low prevalence of cardiovascular disease, which is certainly not attributable to the modest number of AI_M carriers. We put forward the hypothesis that this may be the consequence of a general environmental condition in which the small community lived and which did not allow, up to now, the expression of the selective effect of the mutation.

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